The Digi-Comp II: what the simulation showed

West bay 3: the Digi-Comp II · 1 November 2026

1/2 inch steel balls (12.7 mm, 8.35 g), PLA parts, friction 0.3, no bounce, the board at 15 degrees, 2400 steps a second, the pins' friction 0.4 kg cm^2/s^2, unless a line says otherwise. Each mechanism was worked out alone first, in `tests/`, then on the board, with its tour.

The hand, shown (2026-11-01)

William asked to see the hand: everything the operator does, which until now happened at once and unseen (switches and flip-flops set, START pushed, balls jumped from the store to the top). The hand is now drawn (`parts/hand.scad`: a pointing hand, without a forearm, turned so its palm rises beyond the finger: seen from the front, as the exhibit is, nothing of it hides the fingertip; William asked for both) and touches nothing; each thing it does waits for it to get there. It lifts 9 cm clear of the board, goes over at 60 cm/s, easing off and on, comes down onto the part and does it; with nothing to do it waits beside the board at the right, by the START lever. To wind the weight it goes round the board's right edge to the drum underneath. It visits only what needs changing (`SET A 37` touches the flip-flops that differ), and parts that aren't on the board (PARTS) it doesn't visit. The machine counts as busy while it has something to do, so the programs and their checks wait for it.

Showing it found:

- **R squared the wheel under waiting balls.** Setting the machine back set the wheel square as well as putting the balls back, both at once. With the hand carrying the balls, the balls already waiting stay put, and squaring the wheel by a few degrees under them shoved the column up the INPUT channel and off the board's top. The wheel is now squared only when no ball is in it (which is also when the balls go into its notches).
- **A ball's drawn position lags its body.** The script read where a ball was from its drawing (`obj.pos`), which follows the body only after a step. Set up twice in one instant, the balls all read as lying where they were made, so the hand "carried back" all twelve and stacked them past the top. Balls are read from their bodies now.
- **The balls in the hand count as nothing.** Letting balls go at the top, the hand stacks them on the column waiting in the INPUT channel. The balls it was holding, over the top of that column, counted as part of it, so the new ones went on above the hand, past the board's top, and fell to the floor. And on its way from the store to the top the hand passes over the overflow hole: the hand's watch for a ball there (at any height) saw one, and a multiply read as having overflowed. A ball in the hand is now neither waiting, in the collector, at rest, nor at STOP or OVERFLOW.
- **Refills are counted when picked up.** In AUTO a ball is checked at the trip lever while the hand may be carrying others to the top: counted when let go, a check made meanwhile came up short.
- **The pockets under the plate** (STOP's and OVERFLOW's) can't be reached by a hand: the plate is over them. The hand reaches the hole over each, as you would with a magnet on a stick (a steel ball comes up through the 14.5 mm hole).

With the hand, the whole library (the tour) passed on its own and in the hall, 161 balls right, each program with only its parts. It takes longer: the tour ends at frame 120 880 (98 403 without the hand, 23% more), so `check-all.sh` allows 135 000 frames (135 600 in the hall).

Each program with only its parts (2026-10-31)

William asked for each program to show only the parts it needs, on the full board, and for them to be really absent: out of the simulation, not only out of sight, so that what runs is what a machine built only that far would do (a wrong setting sends a ball off where a missing wall was).

Which part each thing belongs to. Every wall and peg is tagged as it is made in `digicomp2-geometry.lua` (`part`), and the script tags the moving parts and the shutters; `manual/fig-parts.png` is drawn from the same tags. The names are MANUAL section 7's: A, CLEAR, COUNT, MULTIPLY (with D, MQ and STOP), MEMORY, COMPLEMENT, OVERFLOW, AUTO. Some things are needed by either of two parts, and are tagged with both: the lane from MULTIPLY down to CLEAR and the CLEAR switch (a ball for COUNT passes it), and the return lane down the left edge (MEMORY's drops and STOP's). Always there: the board, the collector, the feeder and the MULTIPLY switch, which every ball passes.

OVERFLOW, as the MANUAL had it, was needed to count out and to divide. It isn't, here: watching for the carry out of A7 needs only the overflow hole, and the pocket under it now counts as A's. Without the OVERFLOW shutter the hole is open and the carry drops into the pocket (as HALT); with it, it can also go on to the collector (RUN). So the shutter is only needed to halt the machine in AUTO; `count-out-9`, `divide` and `subtract-12-4` run without it, the subtraction's carry waiting in the pocket.

Taking a part off. A fixed thing (a wall, a peg, a drawn pin, a shutter) is moved far below the floor and back, as the OVERFLOW shutter already is moved. A moving thing (a flip-flop, a flap, the trip lever) is frozen (Bullet's DISABLE_SIMULATION: neither it nor its pin's hinge is simulated), made untouchable (CF_NO_CONTACT_RESPONSE) and moved there too, keeping its setting; put back, it returns where it was. bpp could remove objects and add them again, but the hall runs each exhibit inside its own machinery, and adding and removing over and over has been trouble before (HANDOFF); moving them needs nothing new.

Two parts doing a third's work. Taking parts off found two places where a wall drawn as one part's also does A's work, which the full board had never shown:

- **The clear lane** (lower level). With only A, MULTIPLY, MEMORY and AUTO, 13 x 5 in AUTO stopped after its second ball. That ball, dropping out of A1's done hole, came straight down the lower level and met the collector's wall at x 7.4, slowly, and stopped there, half a centimetre short of the trip channel. Pressed against the wall (it slopes 2.7 degrees), a ball slides along it rather than rolling, and with friction 0.3 it can't, unless it comes with speed. On the full board that ball met the wall at x 5.0 already moving right, steered by the clear lane's walls, and rolled on into the channel. In MANUAL a ball stopped there is still in the collector and harms nothing; in AUTO it trips nothing. So the clear lane is A's as well (and always there); CLEAR is its switch's drop into it.
- **The done lanes' first stretch.** Then 13 x 5 in AUTO went wrong at its sixth ball: A1 turning over carried into A2, set it, and the ball left A2 to the right fast, skipped across A2's long done hole, and rolled on down the plate and off its end. On the full board the done lane's right wall, beside the hole, stops it and it drops. That stretch is A's too, and so is the overflow hole's outer wall beside the hole (it was A's own before OVERFLOW had its shutter); the rest of the lanes are COMPLEMENT's and OVERFLOW's.

So a machine built without CLEAR or COMPLEMENT still needs those walls; `manual/fig-parts.png` shows them as A's.

A part put back where it had been sent. Once the parts worked program by program, the tour (every program in turn) still went wrong in an AUTO program: right after `AUTO` the anchor tripped with no one pushing, and after `RUN` the wheel ran away (on its own: the count-out in AUTO; in the hall: 25 / 5 in AUTO). Logged at the hook, the trip lever read -2.56 radians, far outside its 0 to 0.45, though it had been put back at 0: the thread's length was set too short, and as the lever's pin brought it round the thread held the anchor tripped. The fault was the script's: taking a moving part off kept its pose as `getCenterOfMassTransform()`, which in bpp's Lua is the body's own transform, not a copy, so it followed the body away, and the part was "put back" 5 m under the floor; its pin dragged it up and flung it round. The flip-flops and flaps had it too, but `R` places them by hand as every program starts; the lever nothing places. It keeps a copy now.

The programs. Each has a `PARTS` line; reading a program checks that every step uses a part it names (a `C` with no CLEAR is an error, before it runs). Starting it takes the other parts off and sets the machine up. Run alone, every program in the library passed with only its parts (after the two fixes above, for 13 x 5 in AUTO). The whole library, the tour, each program with only its parts and the board switching between them, passed on its own (161 balls right; it ends at frame 98 403) and in the hall (98 455), the thread hooked in each AUTO program with the lever and the anchor at rest.

Step 2f: the weight and the fly as parts (2026-10-31)

The planned drive (step 2e, "Planned for the next step"), built, with one change: the drum is **under the lower board**, not in the lower level. The plan put it under the plate, in the lower level's 15 mm; there it can't be reached to wind it, and the damper and a ratchet would have to fit in the same 15 mm. Under the board the hand reaches it, and the damper goes on the same axle. The lower level is empty from the wheel to the top edge, and under the board there is 40 cm to the table at the wheel, the stand's feet only at the sides.

The parts. The wheel and the star are glued on a **3 mm steel axle**, 58 mm, turning in 3.5 mm holes through the plate and the board (the wheel's pin pressed into the plate is gone; the wheel's and the star's holes are snug now, for glue). Under the board, from the board down:

- a **ratchet**, 14 mm, twelve teeth, glued on the axle 0.5 mm under the board;
- the **drum**, turning on the axle: a ring round the ratchet with two thin pawls in its teeth, a floor, a barrel 7.5 mm across and 8 mm long for the cord, and a knurled flange to wind it by (printed as two parts, glued: `drum-top` and `spool`, so neither has an overhang);
- a **coupler** glued on the axle's end and pushed onto the damper's shaft;
- the **damper**, Fuji Latex **FRT-G2-200**, screwed (two M2) to the plate of a **bracket** whose legs come up either side to feet screwed under the board.

The cord, 0.5 mm braided line, leaves the drum's left side (forward, the wheel turns clockwise seen from above, so that side moves up the slope), runs 19 cm up the slope under the board to a **pulley** (16 mm groove, on a 3 mm pin in a bracket screwed under the board, reaching 2 cm out past the top edge, so the cup hangs clear behind the board; none of the drive can be seen from in front: the key `W` looks at it from the side) and down to the **cup**, a printed cup (7.6 g) of eight steel balls: 74.4 g in all. Its torque on the wheel is 0.0744 x 981 x 0.4 = 29.2 kg cm^2/s^2 (2.9 mN m), the old `WEIGHT_T`'s 30. Eight balls, not the plan's nine: with the cup's own weight nine would have been 34. From wound up (its bail 1 cm under the pulley) to 1.5 cm over the table it drops 36.1 cm: 57 balls.

The damper's mount. The datasheet's drawing came without its labels: its numbers are 24, 19 +- 0.05, 15, 8, 7, 6.6 (5.4 without the gear), 3.7, 3.6 +- 0.05 (a diameter), 2.6, 2.5 +- 0.05, 2.2 and 2.1. Sugatsune's SRT-G2, the same damper, gives 24 x 15 x 5.4 mm and M2 screws. Read together: a flange

24 mm long with holes 19 mm apart, a body 15 mm across, 5.4 mm high with its shaft, the shaft 3.6 mm across with flats 2.5 mm apart. The bracket's holes and the coupler's socket are drawn to that. **The socket is the one guess that matters:** measure the shaft of the damper in hand before printing the coupler (`DR.DAMP` in `digicomp2-geometry.lua`; the socket is 0.1 mm larger all round).

The ratchet drives the drum's way only: the teeth rise slowly counterclockwise (as seen from above) and drop straight back at a face; the pawls' tips rest on two faces opposite each other. Pulled by the cup, the drum turns clockwise and the pawls push the faces; wound backward, they ride up the slopes and click over.

In the simulation

- **The cup** is a body (the cup and its balls as one: the balls are not loose in it), hanging from where the cord leaves the pulley. It collides with the table only.
- **The cord** is simulated as the thread is: it can't stretch and only pulls. Each step, if it is taut and the cup and the drum would lengthen it, a pull on the cup and a turn of the wheel through the drum's 4 mm radius, just enough to stop that (the cup's weight for the step included). It runs over the pulley without friction, and its run under the board is straight and fixed.
- **The damper** is a drag on the wheel, c times its speed: c = 2 mN m / 2.09 rad/s = 0.955 mN m s (9.55 kg cm^2/s), applied each step so it can't overshoot (`DC_DAMP` scales it).
- **The ratchet** is engaged except while the hand winds: then the drum turns back on its own, the cord pulls on the hand, and the wheel feels nothing.
- **What turns with the wheel** (the axle, ratchet, drum, coupler) adds its inertia, 0.0016 kg cm^2, to the wheel's 0.0148. The cup adds more through the cord: 0.0744 x 0.4^2 = 0.012, nearly the wheel's own. It is in the simulation through the cord, not by adding it to the wheel.
- **The hand winds** (5 cm of cord a second) when fewer than 25 balls are left and nothing has moved for 0.3 s: so never in the middle of an AUTO run (the longest in the library is 21 balls), and the machine is always at rest while it winds.

One thing to get right in the script: the hand counts the wheel as turning above 0.3 rad/s. Held by the anchor, the wheel gets the cord's pull for the coming step (0.74 rad/s for an instant) before the step, in which the anchor's contact takes it back. Measured after the pull, the wheel always looked to be turning, and the first program never began. It is measured before the pull now.

What it does

13 x 5 in AUTO (`multiply-auto.dc2`), the damper as rated: all 21 balls from one START, every one right, A 65. Each quarter turn took 0.40 to 0.49 s, at most 3.3 to 3.8 rad/s (above the 3.06 the cup and damper alone give: the balls on the wheel add their weight), three times briefly 5.6 to 7.1 rad/s. The old torque and cap made every quarter turn 0.52 s.

The whole library (the tour), the weight now kept between programs (setting the machine back, `R`, leaves it where it is, so the hand has to wind it):

The damper	Balls right	Tour ends at frame	Windings	A quarter turn's top speed (rad/s): median, 95%, highest
as rated (c = 0.955 mN m s)	161 of 161	99 343	3	3.65, 4.17, 7.0
0.65 of it (the tolerance's soft end)	161 of 161	97 570	(none: see below)	5.46, 5.72, 7.9
1.35 of it (the stiff end)	161 of 161	100 707	3	2.67, 5.54, 6.2

(The 0.65 run was made before `R` stopped rewinding the weight, so it never needed winding; the weight's lowest was 37 balls left.) The tour of the committed step 2e, with the torque and cap, ended at frame 98 292. In the hall, the tour at the rating passed too.

The anchor comes home slowly now. Each quarter turn is two eighths: pushed, the anchor's first finger lets a pin go and the wheel turns an eighth to the second finger (0.08 s); let go, the anchor's counterweight takes it home, the second finger slides off its pin, and the wheel turns the second eighth. Sliding off, the pin has to creep along the finger, the wheel turning forward a little, driven by a small part of the forces there; the damper resists that creep. With the torque and cap (full torque up to 3 rad/s, nothing resisting below it) the anchor was home within half a second (step 2d). With the damper it is usually quick, but not always: from the push to home, at its rating 0.18 s for half the balls, 1.58 s for the slowest twentieth, 1.84 s at most; at 1.35 of it 0.34, 2.95 and 3.58 s. The hand had waited at most 3 s for the anchor before taking the machine to be still: at 1.35 it read the accumulator while the wheel was still creeping, before the ball was out, and from there every reading was one ball behind (6 wrong, in two programs). It waits up to 8 s now, and the table's 1.35 row is with that. On the machine this is what a stiff damper costs: a slower second half of every turn, in AUTO too. It is also why the damper should not be stiffer than the rating's upper end, and why the anchor's fingers want rollers (they have them): with friction there, the creep could stop.

At the soft end the wheel's top speeds, 4.8 to 7.9 rad/s (46 to 75 rpm), are past the damper's 50 rpm. At its rating the wheel is under 50 rpm (5.2 rad/s) but for a few instants (7 of 154 turns): a ball dropping into a notch kicks it. A damper at the soft end would want a ball out of the cup (seven), or the -300 (0.3 N cm).

Still open

- The datasheet's damping is rated at one speed: linear in speed is assumed. It lists 10 cycles a minute at most, written for lids; here it turns steadily, a quarter turn a ball: to ask the maker.
- The damper's shaft (above), and the bracket's hole spacing, from our reading of the drawing.
- The ratchet's pawls are thin (0.8 mm) and printed in place: if they break, the drum turns free and the cup drops. A wire spring pawl would be sturdier.
- The cord is taken not to stretch, and the pulley to have no friction.

Step 2e: OVERFLOW's halt (2026-10-31)

E.S.R.'s OVERFLOW switch has two settings: RUN, the carry out of A7 goes on and the machine runs on; HALT, it stops automatic operation. Here it is a **shutter** over the overflow hole, as COMPLEMENT's over the done holes, sliding up and down the slope in a groove, its knob up the slope and to the left of the hole, where no ball goes. RUN (closed): the ball rolls over the hole, down a lane (the hole's outer wall, carried on to the plate's end) and off the plate onto the lower level, and on to the collector and the trip lever. HALT (open): it drops into a pocket on the lower level, three walls under the hole, as STOP's, and trips nothing. The hand watches for a ball passing the hole either way.

Counting out nine balls in AUTO (`count-out-auto.dc2`) ran 9 balls from one START, the 9th into the pocket, and A read 0. Dividing 25 / 5 in AUTO (`divide-auto.dc2`) ran 19 balls: four cycles of four, and three balls of the fifth, the third of them (its 4s) going over 128 into the pocket; A read 0 and the MQ 2 (5 cycles begun).

A halt stops wherever in the cycle the overflow comes, so the rest of the divisor is not added. A = 0 there means it went exactly only if that was the cycle's last bit of M: 11 / 5 overflows on its third cycle's 1s, with A = 0 and its 4s not yet added (quotient 2, remainder 1, not 3). So the manual says to finish the cycle by

hand, in MANUAL, unless A = 0 at the cycle's end; the board's MANUAL rule (finish the cycle, the quotient one less, the remainder M - A) still holds after it.

Planned for the next step: the weight and a rotary damper for the fly

(Built in step 2f, above, with the drum under the board.)

The simulation's weight is a torque on the wheel (`WEIGHT_T`, 30 kg cm^2/s^2 = 3.0 mN m) and its fly a cap on the wheel's speed (`FLY_W`, 3 rad/s, about 29 rpm). As parts:

- **The weight:** a cup of nine spare balls, about 76 g (the cup's own mass to be added), on a cord wound on a drum of radius 4 mm on the wheel's shaft, under the plate (the anchor's counterweight takes the room above the wheel), the cord over a pulley at the board's top edge. Torque 0.076 kg x 9.81 m/s^2 x 0.004 m = 3.0 mN m. One ball out is a quarter turn of the wheel: 6.3 mm of cord, so ~30 balls is a drop of ~19 cm (the drop the stand allows sets how often it is wound). Rewinding goes through a ratchet between the drum and the shaft, so winding turns neither the wheel nor the damper. Each ball in the cup is ~11% of the torque: the cup is how the speed is trimmed.
- **An air fly does not suit:** to hold 3 rad/s a vane of a size that fits would have to turn ~200 times as fast as the wheel, a gear train of a music box's.
- **The fly: a rotary damper**, bought: **Fuji Latex FRT-G2-200** (also sold by ACE; RS 793-2945 for the -G1 with a gear). Rated 0.2 +- 0.07 N cm (2 +- 0.7 mN m) at 20 rpm, 23 degC; both ways; silicone oil in a polycarbonate body, a 15 mm flange, acetal shaft; 0.6 g (0.8 g with the G1 gear: module 0.5, 14 teeth, 7 mm pitch circle); 50 rpm at most; 0-50 degC (colder, stiffer). The larger sizes of the series are -300, -450, -600 and -101 (0.3 to 1 N cm), the FRT-C2 series 2 to 3 N cm: too stiff here, ~7 to 10 times. Mounted on the wheel's shaft, 1:1.
- **In the simulation:** a drag torque c w in place of the cap, with c = 2 mN m / 2.09 rad/s = 0.96 mN m s; with 3.0 mN m of weight the wheel settles at 3.1 rad/s. The tolerance (+-35%) gives 2.3 to 4.8 rad/s: the library to be run at both ends.
- **Open:** the datasheet says only that torque rises with speed (linear is assumed); it lists 10 cycles a minute at most, written for lids, and the wheel turns steadily: to ask the maker. Mount sizes are to be taken from the datasheet drawing (the copy fetched here was garbled).

Sources: Fuji Latex, FRT-G2 datasheet, https://www.fujilatex.co.jp/en/wp/wp-content/uploads/2017/01/FRT-G2.pdf ; FRT-C2/FRN-C2, https://www.fujilatex.co.jp/en/wp/wp-content/uploads/2017/01/FRT_FRN-C2.pdf ; RS, https://uk.rs-online.com/web/p/rotary-dampers/7932945 .

Step 2d: AUTO (2026-10-28)

The escapement on the board

The feeder's knob is gone: the wheel is turned by the weight (a torque of 30 kg cm^2/s^2 on its pin, `DC_WEIGHT`) no faster than the fly lets it (3 radians a second, `DC_FLY`), and held by the anchor, as in `tests/escapement.lua` (step 2a, below). The hand's START is a push on the anchor's arm for 0.3 s (`DC_PUSH`, 300). On the board it needed:

- **The anchor's stop posts in the simulation** (they had been drawn only): without them the anchor, pushed, swung on to 90 degrees.
- **Waiting for the anchor to come home.** A finger sliding off a pin brings the anchor back slowly, up to half a second; the simulation had come to rest before it was home, and the next ball came one push late. The machine now counts as busy until the anchor is within 1.5 degrees of rest (or 3 s have passed).

- **Rollers** on the fingers: the anchor's friction is 0, as the test found.

With that the whole library (all MANUAL) passed as before.

The trip

The collector's wall now stops short of the right-hand side, and every ball turns down a **trip channel** there, under the trip lever, and on to the **store** below it (a second wall, sloping down to the left, where the balls wait for the hand). Getting a ball into the channel took three tries:

- The wall ended 2 mm short of the channel's left wall: the balls went down that gap, outside the channel, past the lever.
- It ended over the channel, at its middle: the balls stuck in the corner between its end and the rim. Past its end a ball needs its whole width clear to the rim (the end and the rim were 7.6 mm apart).
- It ends on the channel's left wall: 14.7 mm to the rim, and every ball turns down the channel.

The lever. The first lever's arm reached up the slope from its axle, its end just above a ball's middle, and its post pulled a thread to a crank on the anchor. A ball arriving under the arm's end never lifted it: it met the end's corner 1.9 mm over its middle, so the push was 17 degrees above the board, and below the line from the corner to the axle (34 degrees). It only pushed the arm down onto its stop, and the ball stopped. The arm now **trails down the slope** from the axle: a ball rolls under the axle and meets the arm's underside, which it can only lift. A post up from the axle then tips *up* the slope as the arm lifts, so it **pushes**: a rod, not a thread, to the anchor's counterweight arm, extended past the ball, 35 mm from the pin. Pushed up the slope there, the anchor trips. (The rod is gone now: the lever pulls a thread, try 6 below.)

Measured on the board, with the rod nearly weightless (0.3 g, the first try): a ball lifts the lever 0.34 to 0.35 radians (0.38 once); through the rod that takes the anchor from -7.9 to +5.0 degrees of its -7.9 to +7.9 degree swing, 82% of it. That was enough: the second finger catches the next pin, and every trip let out exactly one ball.

The rod's weight. That was with the rod nearly weightless (0.3 g). A real rod 1.5 m long weighs more: a 1.5 mm carbon-fibre rod, about 4 g. Its weight down the slope pushes the post back down the slope, against the ball: with a rod of 1, 2 or 3 g the first ball no longer lifted the lever enough, and the machine stopped after it. So the rod's weight has to be balanced, and that took five tries, and then a sixth design without a rod at all:

1. **A counterweight on the lever, up the slope over its axle** (a steel ball on a short arm). AUTO then passed with any rod from 0.3 to 8 g. But MANUAL took the rod off, and without the rod's weight the lever tipped over into the channel, its counterweight down among the balls, and the 13 x 3 stopped at its 13th ball, all twelve balls behind it.
2. **The rod on for good,** working an **AUTO crank**: an arm of its own on the anchor's pin, over the anchor and its counterweight ball, as long as the anchor's counterweight arm. AUTO is a latch pin through the crank's end and the end of that arm; MANUAL, the pin out, leaves the crank swinging idle. MANUAL still failed: a counterweight over the axle balances the rod only near rest. Past 0.14 radians it is top-heavy, so once a ball had lifted the lever it fell on over and stayed there, counterweight in the channel. (The lever's post had also gone up to the crank's height, 37 mm over the plate; then a 2 g rod failed in AUTO too, the counterweight, made for 4 g, holding the anchor half tripped. The taller the post, the more any mismatch between rod and counterweight turns the lever; the post is short again, its top 21 mm over the plate, and the rod slopes up to the crank, 0.7 degrees.)
3. **A counterweight on the AUTO crank**, across its pin (in the board's plane, so the same at every angle): MANUAL passed, but in AUTO a ball stopped under the lever, and the machine with it.

4. **A thread instead of the rod,** from the post down the slope round a pulley and back up the board (nothing to balance: a thread weighs next to nothing): in the simulation the closed loop of lever, thread and pulley held the lever down, and the first ball stopped under it, though a motor on the lever's axle turned it easily.
5. **A counterweight hanging below the lever's axle,** on an arm glued on the axle's end where it comes out beyond the channel's left wall, behind the collector's wall, where no ball goes: a 3/8 inch steel ball (3.53 g), 3 mm below the axle and 6.4 mm up the slope, which balances a 4 g rod. Below the axle it is steady at every angle. (Hung lower, 9 mm below, it was too steady: it pressed the lever down on the first ball, which stopped under it, the anchor half tripped, and the wheel ran.) This passed, in MANUAL and AUTO: 13 x 5 in AUTO, all 21 balls from one START, with a rod of 0.3, 2, 4 and 8 g, and at friction 0.2, 0.3, 0.45 and 0.6.
6. **A thread, pulling (2026-10-28).** William's point: a rod 1.5 m long and that thin can pull but hardly push (a 1.5 mm carbon rod buckles at about 14 g; the trip needs about 2, but only while the rod stays straight). A thread can't push at all, so the lever's part that it is tied to must move *away* from the anchor as a ball lifts it: an arm hanging down from the axle, whose foot swings down the slope. The axle comes out through the rim, the arm hangs beside the board, and the thread runs straight up beside the board's right edge, outside everything, to an eye level with the START lever, and across to the START lever, 19 mm from the anchor's pin. It pulls the lever to the right, as the hand does. A thread weighs next to nothing, so there is nothing to balance: the counterweights, the AUTO crank and the latch are gone. In the simulation the thread can't stretch and only pulls: each step, if it is taut and the lever and anchor would lengthen it, an equal pull on both ends stops that (as a spring of nylon's stiffness, the ball bounced off the lever and stopped). Earlier, a thread run round a pulley at the bottom, its two pieces simulated as rigid links, had held the lever down.

This needed the lever's underside changed. Its straight underside met a ball 3 mm from the axle, where the ball has little leverage, and now and then a ball stopped there, under the axle (the 13 x 3 in MANUAL, its 13th ball; it also explains some of the failures above). The underside now stays high out to 6 mm from the axle and then slopes down to the end: a ball meets it farther out. Then 13 x 3 in MANUAL and 13 x 5 in AUTO passed, the AUTO one at friction 0.2, 0.3, 0.45 and 0.6, the MANUAL one at 0.3 and 0.6.

Friction. At friction 0.6 (the slider's top) a ball once stuck under the lever (with try 2): wedged between the floor and the arm, it can't roll on both, and slid too stiffly on the arm. The lever stopped a third of the way up, holding the anchor half tripped, where neither finger holds a pin, and the wheel let nine balls go at once. So the lever's underside is faced with PTFE tape (0.1 on steel, whatever the slider says).

STOP drops its ball into a pocket under the plate, beside the return lane: it never reaches the lever, nothing trips, and the machine stops.

Checking it

In AUTO the next ball is already let go when a ball lifts the lever, so the program's `RUN` checks each ball there (everything it changed, and that it reached the bottom), with one ball in flight; the ball through STOP is checked when all is still. `programs/multiply-auto.dc2`, 13 x 5, ran 21 balls from one START, every one right, and A read 65. The hand keeps the input filled from the store as it runs.

Still to do

- OVERFLOW halting the machine, for dividing and counting out in AUTO (a flap under the OVERFLOW hole, and a pocket, as STOP's).
- The weight, its winding and the fly as parts.

- The thread as it is: it stretches a little and rubs in its eye; in the simulation it does neither, and weighs nothing.

Step 2c: the complement, subtracting and dividing

Done lanes and shutters

E.S.R.'s machine complements with six mode flip-flops between A's (T1 to T6) and two more to sequence them (CF1, CF2). Here the same thing is done with fewer moving parts: under each of A1 to A6's done holes a **done lane** runs on into the next flip-flop's funnel, and a **shutter** covers the hole. Closed, a ball leaving a flip-flop on its "done" side rolls over the hole and on into the next one, as a carry does, so it turns over every flip-flop to A7. Open, the hole takes the ball as before. All six shutters are on one bar: the COMPLEMENT switch.

- **The lanes needed room.** A done lane runs from the hole (3.1 cm right of its flip-flop, 1.3 down) 4 cm across to the next funnel's long right wall, and no lane may run flatter than 35 degrees. So the accumulator's steps went from 8 cm down the slope to 14.75 (`G.ADZ`; the distributor's and MQ's stay 8): the board went from 1216 to 1573 mm long.
- **The carry drifts.** Falling 14.75 cm instead of 8, a carry (which leaves its flip-flop moving left) drifted out past the short left wall of A5's funnel: on the board, the complement of 25 stopped at A4 (25 -> 22, the ball lost). Every one of A's funnels now has both walls long, as A1 to A4's had already (for the memory's lanes); then it went right. Traced, the carries into A5 to A7 still reached the very end of that long wall (4.5 cm left of where they left), and at 1200 steps a second one went past it. So A5 to A7's long left walls go on straight up the slope as **guides**, to the level of the flip-flop above. (A2 to A4 have the memory's lanes there, whose walls do the same.)
- **The hole has no wall below it.** Before, each done hole had a wall just below it that stopped the ball over it. With the lane going on below, there is none: the ball must drop in as it passes. In the test it did, every time, square (14.5 mm) or longer down the slope (20 mm), at friction 0.1 to 0.5, pin friction 0.1 to 3 and slopes 8 to 25 degrees. But on the board tilted 2 degrees across (down to the left), balls rode over the done holes and on down the lanes, adding wrongly: a square hole leaves only 0.9 mm either side of the ball, and a ball running down a little off its middle rides one edge. So the holes are 20 mm down the slope and 18 mm across (`G.DONE_L`, `G.DONE_W`; wider to the left, where nothing stands), 2.7 mm either side; the test then passed tilted 2 degrees either way (3 degrees fails, as everything did before).
- **The shutters slide across, not down.** Sliding down the slope, a shutter leaves its groove open in the lane's floor, a 2 mm pit the ball must cross. Sliding across, to the right, the groove it leaves is under the lane's right wall and outside the lane, and when it is open, the groove left is the hole itself. Each shutter is a 2 mm tile in a groove in the plate's face, flush; a pin stands up from its right end outside the lane, and the bar joins the pins 16 mm over the plate, above the walls (12 mm) and the balls (12.7 mm): nothing that rolls touches it.

`tests/complement.lua` (three flip-flops of A and their lanes, from the exhibit's geometry): thirteen balls, every case of the three with the shutters closed and the ordinary counting cases with them open, each checked for what it turned over and where it went. With the board as it is now, it passed for A1 to A3, A2 to A4, A3 to A5 and A4 to A6, at friction 0.1, 0.3 and 0.5, pin friction 0.1, 0.4 and 3, slopes 8, 15 and 25 degrees, and tilted 1 degree across either way.

Two balls, and the hand

The complement is two balls here: COMPLEMENT on, one ball (127 - A); COMPLEMENT off, one more (+1). The hand moves COMPLEMENT between them, the job CF1 and CF2 do on the original. When AUTO is built, that is the one step of the complement it won't do by itself.

Dividing, and the remainder

E.S.R.'s manual divides by adding the divisor to the dividend's complement until A overflows (OVERFLOW on HALT), the MQ counting the cycles; its chapter on remainders could not be read here. Working it through on this board: a cycle adds M a bit at a time (8s, 1s, 4s, 2s), and A can go over on any of those balls. Stopped at once, the rest of M is never added, and A then says nothing useful: 11 / 5 goes over on the third cycle's 1s ball (127 + 1) and would stop with A = 0, as 10 / 5 does, and read as 3 exactly. Finished to the end of the cycle (its 4s ball too), A = 4. In general, finished, A = cycles x M - dividend, so:

- A = 0: it went exactly, the quotient is the cycles;
- otherwise the quotient is the cycles less one, the remainder M - A.

So the hand watches OVERFLOW and stops at the end of the cycle (when the next ball would go to the MQ). The tour divides 13 by 5 this way: the third cycle goes over (on its 4s ball), A is left at 2, the MQ has counted three cycles: quotient 2, remainder 3.

On the board

The tour (now also the complement of 25, 12 - 4 and 13 / 5, 75 balls in all, every one checked, and the quotient and remainder) passed in every one of these:

- friction 0.2, 0.3, 0.6
- pin friction 0.1, 0.4, 3
- slope 10, 15, 20 degrees
- 1200, 2400, 4800 steps a second
- tilted across 1 degree either way, and 2 degrees down to the right

and failed:

- **tilted 2 degrees down to the left.** Carries that fall the length of the longer step arrive slowly, after the guide and the funnel's long walls, and the tilt carries a slow ball over the apex of a flip-flop at 0 to its far side, past it unturned (traced in `tests/complement.lua`, CP_FROM=3, CP_ROLL=-2). Step 2b passed 2 degrees either way. So: **level across within 1 degree** now.
- **friction 0.1,** once in 75 balls: D3, the distributor's, left the previous ball's swing at 5 degrees of its 12 (the ball slid out of its cup before the swing was done, and the pin's friction held it there), and the next ball read it the other way. The distributor is as it was in 2b (step 2b's own tour, run again now at friction 0.1, fails the same way, on its 17th ball, with D3 at 5 and 6 degrees after several balls. Its list below says it passed; it doesn't now. The narrated tour, since, changed the timing between balls, which may be why; not looked into further). So: friction 0.2 to 0.6. A stronger over-centre pull on the rockers would be the cure; not tried.

The printed parts

Every STL is one watertight solid (`make-stls.sh`: 18 upper and 18 lower tiles now, 214 x 182 mm at most; the shutter; the bar's knob). Put where they are at rest, COMPLEMENT off, no two parts overlap but each foot's stem in its rocker's socket, as before: the shutters lie in their grooves, the bar clears every wall, peg and rocker.

Step 2b: adding and multiplying

Laying it out

Thirteen flip-flops, seven switches and the lanes between them have to fit on one level, and lanes cannot cross there (a ball can go down to the lower level, never back up). The board was laid out in a plan drawn from the geometry (every wall, hole and peg, the lanes coloured by kind) and changed until nothing touched. These settled most of it:

- **The order of the memory.** The memory's switches stand beside A1 to A4, down the accumulator's staircase, so the lanes reaching them, from above and the left, must keep their order: the leftmost of the distributor's four outlets can only go to the lowest, M4. So D2's left goes to the MQ and on to M4, D2's right to M3, D3's left to M2, D3's right to M1, and with the distributor set to start, the balls of a cycle go to the MQ (and M4), M1, M3, M2. The manual's order after the MQ is M2, M3, M1; the sum is the same.
- **The distributor is a tree,** not the original's ring: D1 sends the balls to D2 and D3 by turns, each of them by turns to its two sides. D2 is where the next step of a staircase would be, so D1's left drops straight into its funnel, and D2's left straight into MQ1's; D1's right, and every other outlet, falls into a **catch** (a funnel where the next flip-flop's would be) and runs on down a lane. D3 must stand far enough from D2 that D2's right catch and D3's left don't overlap (8 cm right of D1), and low enough that the lane to it falls steeply enough (14.2 cm below D1).
- **Lanes** fall at least 35 degrees from across the board (the geometry warns of any that don't); the flattest, D1's to D3, falls 39 degrees. The longest are MULTIPLY's down to CLEAR and D2's to M3, 27 cm each.

The board grew from 322 x 878 to 424 x 1216 mm (14 tiles a level), and its table from 100 x 150 to 110 x 180 cm.

Two ways into one funnel

A1 to A4 each take balls from two places: the carry from the flip-flop above (or COUNT, for A1) and the memory. The carry was traced first (a copy of `tests/toggle.lua` printing the ball's centre every 1/60 s): it leaves the cup moving 11 cm/s across and crosses the next funnel's mouth 4 mm left of its centre, so its left half is not free for a lane ending over it. Instead each of these funnels has a **long left wall**, out 35 mm and up 15.7 mm (31 degrees), and the memory's outlet ends over that wall, 26 mm left of the throat: its ball drops onto the wall and rolls down it into the throat, beside the carry's path, not across it (`tests/junction.lua`: the carry and the memory's balls in turn, nine of nine right).

- **Under the lane's end.** With the outlet ending 19.7 mm above the throat, the memory's ball stuck at once: rolling down the long wall, it had to pass under the end of the outlet's own right-hand wall, which left 8 mm. It ends 28 mm above the throat now. The MQ's done balls join the same way (a long right wall under MQ2's and MQ3's catches, mirrored), and stuck there the same way, first.
- **COUNT** at first ended over the right half of A1's funnel; its ball stuck under its own outlet's left wall, as the memory's had. Centred over the throat, it worked, but with the pins' friction at 2 or 3 A1 stopped near the middle (+4 to +6 degrees) after the first ball, and the next went the wrong way: the ball has run 40 cm down from MULTIPLY, and came into the top of A1's run at 34 cm/s (the toggle test's ball starts there at rest). A1's funnel has a long right wall too now, and COUNT's outlet ends over it as the memory's do over the left: the ball drops onto the wall and rolls in slowly.
- **The clear lane** starts at CLEAR's drop, right of COUNT now, and bends left at once to pass where COUNT's off balls land. Coming down straight to A1's foot, the clear ball was pushed by the narrowing past its line into A1's wedge (at 0), and with the pins' friction at 0.1 its rubbing dragged A1

over to 1. The lane now overshoots to the left and comes back a little to the right, so the ball meets the foot running along the right wall, as at every other foot.

The memory's switches

The same switches as CLEAR and COUNT, but a 0 has to drop the ball before it can reach the next switch's inlet, only 5 cm below: so the "0" outlet ends at once in a hole just under the flap's pin, its walls round it. The first one stood the hole's inner wall where the divider's slope begins: the ball rested on its end, beside the flap. The wall now begins where the divider's slope ends. A ball dropped there rolls down a wall on the lower level (40 degrees) to the **return lane** down the left edge, and on to the collector, away from the clear lane.

The MQ

Three flip-flops down a staircase like the accumulator's, read the other way: a ball that finds one at 1 (apex left, as it is lettered) sets it to 0 and leaves to the right (done); one that finds it at 0 sets it to 1 and goes on to the next (the borrow). So each ball takes one away, the same parts as the accumulator adding one. A borrow out of MQ3, when the MQ was 0, drops through STOP: the hand watches it (as you watch the machine in MANUAL) and turns no more. The done balls of MQ1, MQ2 and MQ3 run together through the catches under MQ2 and MQ3 (each takes the lane from the one before on its long right wall) and on to M4.

On the board

The tour (README.md): COUNT 0 to 2, CLEAR, ADD 37 + 11 (five balls: 48, then STOP), CLEAR, MULTIPLY 13 x 3 (thirteen balls: 39, then STOP), twenty-two balls. After each ball it checks the accumulator, the MQ, the three distributor flip-flops and STOP against what one more ball should make of them, and that the ball reached the collector. The hand puts balls from the collector back into the input when fewer than four are waiting. It passed in every one of these:

- friction 0.1, 0.3, 0.6
- pin friction 0.1, 0.4, 1, 2, 3
- slope 12, 15, 20, 25 degrees
- tilted across 1 and 2 degrees either way
- 1200, 2400, 4800 steps a second
- the hand's torque on the knob 30, 50, 1000 kg cm^2/s^2

(eighteen runs, each the whole tour). Before the two changes to A1 above (COUNT onto a long wall, and the clear lane's bend), pin friction 0.1, 2 and 3 failed there; the others tried then (friction 0.1 and 0.6, 1200 and 4800 steps, the hand's torque 30 and 1000, slope 25, tilted 2 degrees) passed. `tests/junction.lua` passed (nine of nine) at friction 0.1 and 0.6, pin friction 0.1 and 3, and slopes 12 and 25 degrees.

The computing was right the first time the balls could get through: the distributor, the MQ and the memory gave 48 and 39 once the three places above (the two funnels and the memory's hole) let the balls by.

The printed parts

Every STL (`make-stls.sh`: 14 upper and 14 lower tiles now, 212 mm across) is one watertight solid. Put where they are at rest, no two parts overlap but each foot's stem in its rocker's socket (0.3 mm^3, where they are glued), for all thirteen flip-flops and seven flaps.

Showing where the ball will go

Before each ball the exhibit works out, from what the rockers and switches read, the way the ball will take and what it will do, draws the way as a yellow dashed line on the board (360 short dashes, all alike, moved into place) and says it on the sign (THIS BALL). It is the same reckoning the tour checks each ball against (the accumulator, the MQ, the distributor and STOP after every ball), so a tour that passes did what the line and the sign said, ball by ball.

What it costs

About 430 fixed boxes (320 walls, the plate round its holes, the pegs and posts), twelve balls, thirteen rockers, seven flaps and the wheel. The tour, about 215 simulated seconds (each ball's way is shown for 1.5 s before it goes), took about four minutes of computing headless here (twice the rockers and twice the length of 2a's); between balls, and when it is done, the exhibit rests and costs nearly nothing.

Step 2a: the accumulator

Two levels

E.S.R.'s board has two levels, and so has this one: the upper plate (4 mm) on which the balls compute, and under it the lower level, 15.2 mm high (a ball and 2.5 mm), which takes every ball that is done down to the collector, under everything else. It is what makes a seven-bit accumulator fit: step 1's done lanes each needed a strip of board down to the bins, and seven of them would not have fitted beside the staircase.

- **Done holes.** A ball tipped out of a rocker's right cup drops through a 14.5 mm hole 31 mm right of and 13 mm below the pin, a wall beyond it and below it so it can't run past. With the hole 1.5 mm further out, the flip-flop's test failed at friction 0.6 and at pin friction 3.
- **The collector's wall** is past the end of the plate, with nothing over it. At 15.2 mm high, a ball that had rolled the length of the lower level at a slope of 22 degrees or more jumped it and left the board. It is 30 mm high.
- **The collector** first sloped down to the left: the balls lined up from the left end, over the end of the clear lane, and the ninth ball blocked it, so the tenth stopped in the lane. It slopes down to the right now (2 cm across the board), away from the clear lane and the overflow, and the clear lane ends 35 mm short of its wall, so a ball coming out of it can turn and roll along the wall (ending 10 mm short, two balls were trapped between the lane's end and the wall).
- **The input channel** holds ten balls (13.5 cm); a ball placed past its end fell off the board.

The flip-flop's foot: CLEAR

The original clears its accumulator with a ball on the lower level. Here each rocker has a **foot** on the lower level, joined to it by a stem through a square hole in the plate: an arm 20 mm up the slope from the pin (down the slope, its weight would have made the rocker hang like a pendulum instead of falling over to one side: tried first, it did), ending in a wedge that leans right when the rocker reads 1. A ball coming down the **clear lane** beside it meets the wedge's sloping face and pushes it left, which is the way that turns the rocker to 0. With the foot, the whole flip-flop weighs 5.45 g (rocker 2.5 g, foot 1.9 g, steel ball 1.0 g), its centre of mass 4.4 mm up the slope from the pin: the foot up the slope makes its two leanings deeper than step 1's rocker's (2.4 mm).

- **Pushed part way.** At first the lane was straight, and the ball pushed the wedge only until it could squeeze past it. That left the rocker a little past the middle, where its weight hardly turns it and the

pin's friction holds it: at friction 0.6 it stopped at +3 degrees, still reading 1, four times in eleven. And the lane had to be placed within a millimetre either way (13.5 to 15.5 mm right of the pin both failed).

- **The narrowing.** Now the lane's right wall comes in beside the foot, so the ball passes the wedge with its left side 0.5 mm clear of where the wedge is when the rocker is at 0 against its stop. A ball that meets a wedge at 1 must push it all the way back to 0 (the rocker then reads -11 degrees); one that finds it at 0 touches nothing. It works with that gap anywhere from 0.2 to 1.5 mm, at friction 0.1 to 0.6 and pin friction 0 to 4 (`tests/toggle.lua`, eleven or sixteen balls each, from above and down the clear lane in turn), and at slopes of 8 to 25 degrees.
- **Corners.** Two places jammed a clear ball, both found with `DC_CONTACTS`: the lane's left wall coming back just where the right wall narrowed (leaving 12.4 mm for a 12.7 mm ball), and the narrowing starting above the bend where the lane from the flip-flop before joins it (its corner stood in the lane). The left wall now stays open until past the narrowing, and the narrowing starts below the bend. A lane's walls are mitred at its bends.

The switches

A flap 20 mm long and 3 mm thick on a pin at the tip of the divider between two outlets, its free end against one wall of the inlet. It was right ten times in ten (set by hand left and right in turn, then a ball) at friction 0.1 and 0.6, pin friction 0 and 2, slopes 8 and 25 degrees, and with the ball run in from 1 or 15 cm above (`tests/switch.lua`). The divider's tip must stay 5.5 mm below the flap's pin: at 1.5 mm the flap jammed against it.

AUTO: the START lever (the escapement works; not on the board yet)

The original's AUTO lets a ball reaching the bottom release the next from the top, through the START lever. A long lever on the lower level was tried, its top end carrying two teeth up through slots into the input channel (an escapement: one ball out each time the lever is pushed) and its bottom end a wedge in a trip lane. It failed: round teeth coming in from the sides of the channel are pushed sideways by the balls they hold, and with the lever's pin 9 cm away that was enough to hold the lever pushed (it never came back after the hand let go), and a trip ball let all six balls out at once. The next design needs teeth whose faces meet the balls square on, on a lever whose pin is in line with the channel, so the balls' weight turns it neither way.

Second attempt (2026-10-27): the wheel held by an anchor. The feeder wheel itself as the escapement: let the balls' weight turn it, and hold it a quarter turn at a time with a clock's anchor (two fingers on a pin to the wheel's right, catching four pins on an escape star above the balls). Each finger lay along the line its pin moves on, through the anchor's pin, so the wheel's push could not turn the anchor. In a test of the wheel and anchor alone, the anchor (a 1/2 inch steel ball its counterweight) swung to tripped when pushed and back by itself when let go (not quite home: against a pin that had stopped short). But the wheel would not turn:

- **Balls coming down onto the top of the wheel** (as now): the next ball in the channel dips into the notch as the ball in it is carried away, and the notch's edge must then push it, and every ball above it, back up the channel. The hand's knob can (it is allowed 50 kg cm^2/s^2); the weight of one ball in the wheel gives about 3, and the wheel stopped 12 degrees round.
- **A paddle wheel fed from the side,** the channel coming down in line with its left-hand space so the balls' weight pulled it round: the next ball still follows the one being carried away, and the vane coming round next must sweep through the channel and push it back. It stopped 58 degrees round, a ball jammed between the vane and the channel's wall.

Whatever lets one ball go must at some point push the next back against the balls above it. So that push has to come from somewhere other than the balls in the wheel: from the trip (a ball arriving at the bottom has fallen over a metre, but loses most of that at the collector's wall), or, more surely, from a weight the hand winds up, as a clock's, turning the wheel through the anchor, which already worked. That is the next thing to try.

Third attempt (2026-10-27): a wound weight, a fly, and straight fingers. The wheel driven by a steady torque (the weight) with a top speed (a fly, the air brake on a clock's striking train), and held by the anchor (digicomp2-geometry.lua, G.ANCHOR; tests/escapement.lua). It works: at rest nothing comes out, and each trip, from a tenth of a second to two seconds long, lets out exactly one ball and the anchor comes back home. What it took:

- **Each finger on its own side of the pin it stops.** The first finger stops a pin coming toward the anchor's pin, and lies on the near side of the stopping point; the second stops a pin moving away, and must lie beyond it, on a bar carried over the pins. With both on the near side the second never touched its pin.
- **Straight fingers, not curved ones.** Fingers curved as a deadbeat clock's (arcs about the anchor's pin, so the pin slides along them while the anchor swings) held the anchor tripped whenever the wheel was driven: the pin caught on the arc's edge, and the push there no longer passed through the anchor's pin (it turned the anchor half as hard as the counterweight, the other way). Straight fingers along the line through the anchor's pin have no such edge.
- **Rollers on the fingers** (no friction at the anchor): it passed with friction 0, 0.05 and 0.1 there; with the curved fingers any friction held it.
- **A weight big enough to push the column back.** At the board's 15 degrees, a torque of 20 to 60 kg cm^2/s^2 on the wheel works and 10 does not (the wheel can't push the next ball back up the channel); at 25 degrees 30 is too little for a short trip, and 50 works.
- **A fly.** With the top speed 1.5 to 12 radians a second (a quarter turn in a fifth of a second to a second) every trip let out one ball; with no limit one trip in eight let none.

It passed (eight trips each) with friction 0.1, 0.3, 0.5; pin friction 0.1, 0.4, 2; slope 8, 15, 20 degrees; 4 and 9 balls in the channel; the hand's push 100 and 300.

Still to do before AUTO is on the board: the weight, its winding and the fly as parts (here a torque and a speed limit on the wheel's pin); the trip, a ball arriving at the bottom pushing the anchor's arm across; STOP, whose ball must not; and the AUTO/MANUAL switch between the hand's knob and the weight.

The accumulator on the board

The tour (README.md): COUNT from 0 to 4; 125 to 127 and over to 0, the carry running through all seven; CLEAR from 90 and from 127; CLEAR at 0; both switches off. Every reading and every ball's arrival in the collector checked, eleven balls. It passed in every one of these:

- friction 0.1, 0.2, 0.3, 0.4, 0.5, 0.6
- pin friction 0.1, 0.4, 1, 2, 3 (a quarter of the estimate to seven times it)
- slope 8, 10, 15, 20, 25 degrees (at 8 the tour takes longer)
- tilted across 1 and 2 degrees either way
- 1200, 2400, 4800 steps a second
- the hand's torque on the knob 30, 50, 1000 kg cm^2/s^2

and failed with no pin friction at all (the 1s rocker bounced back off its stop, on the board though not in its test; any real pin has some), and tilted 3 degrees across either way (as in step 1, the balls are carried past the apexes). So: level across within 2 degrees.

The printed parts

Every STL (`make-stls.sh`) is one watertight solid. Put where they are at rest (`PART="rest-..."` in `parts/digicomp2.scad`), no two parts overlap but each foot's stem in its rocker's socket (0.3 mm^3, where they are glued). The bottom lower tiles first came out with the collector's wall cut off at the plate's height, and the collector's tall end walls overlapped the plate's rim; both fixed.

What it costs

A few hundred fixed boxes (walls, the plate round its holes, pegs, posts), twelve balls, seven rockers, two flaps and the wheel. The tour, about 90 simulated seconds, took about a minute of computing headless here; the exhibit rests (stops stepping) whenever nothing has moved for half a second, so between balls it costs nearly nothing.

Step 1: the counter (2026-10-26)

Three flip-flops on a one-level board, with lanes to bins for the done balls. The rocker then had no foot, and `tests/toggle.lua` was the rocker alone under its channel (it is now the rocker with its foot: see step 2a).

1/2 inch steel balls (12.7 mm, 8.35 g), PLA parts, friction 0.3, no bounce, the board at 15 degrees, 2400 steps a second, the pins' friction 0.4 kg cm^2/s^2, unless a line says otherwise. The flip-flop was worked out alone first, with `tests/toggle.lua` (one rocker under its channel, balls let go one at a time, 16 of them), then in the counter, with its tour (eight balls from 0, each reading checked).

The flip-flop

The idea. A rocker on a pin whose centre of mass is up the slope from the pin is over-centre: it falls to one side or the other against a stop and stays, a bit of memory. Its top is a roof whose apex is up the slope from the pin, so the apex leans with it. A ball coming straight down meets the roof beside the apex, on the side that is raised, and its weight on that side takes the rocker over. So each ball goes the other way from the last. The first rocker, an inverted V of plain strips, did this sixteen times out of sixteen at every slope from 6 to 30 degrees and every friction from 0.1 to 0.6, as long as it leaned at least 8 degrees (at 6 degrees the apex moved too little to choose a side for the ball).

Cups. With friction at its pin, the plain roof stopped half way over: the ball runs off the end of the roof while the rocker is still near the middle, where its own weight hardly turns it, and the pin's friction holds it there. The next ball then finds no raised side. At 0.8 kg cm^2/s^2 of pin friction, 14 of 16 balls went the wrong way. A **lip** at each end of the roof makes a cup, which holds the ball until the rocker has gone over; at the end of the swing the cup tips it out. With 5 mm lips the rocker worked at four times that friction. The lips lean out 67 degrees from straight up the slope:

Lips (lean out)	Counter's tour, under 10 hard conditions
none	fails at 0.8 kg cm^2/s^2 of pin friction (the rocker alone)
60 degrees	8 of 10: a ball stays in a cup when the table is tilted 2 degrees across
67 degrees	9 of 10: with no pin friction at all, a rocker bounces back from its stop
75 degrees	9 of 10: the same
3.5 mm long, 67 degrees	8 of 10: too short to hold the ball over at high pin friction

(The ten: tilted 1 and 2 degrees across either way, pin friction 0 and 2, friction 0.1 and 0.6, slope 10 and 20 degrees. These were run with the pin friction first estimated, 0.25 kg cm^2/s^2, before the steel ball

below, which these rockers did not have.)

The weight. The bouncing back is a sign that the rocker's two leanings are too shallow: its centre of mass was only 1.5 mm up the slope from the pin, so the dip it rests in is tiny, and a rocker coming hard onto its stop can rebound right out of it. A **steel ball of 1/4 inch (1.04 g)** pressed into the rocker under its apex, 5.8 mm up from the pin, moves the centre of mass to 2.4 mm and makes the dip deep enough: with it and the 67 degree lips, the counter passed all ten (and so did 1 and 2 g, 5 mm up, tried before it).

The apex. The first rockers were built of boxes, and the strut from the pin to the apex stood 3.5 mm above the roof: the ball landed on its corner and went either way. The rocker is now built of convex pieces whose upper edges are the roof's lines exactly, nothing above them (and the printed part is the same pieces).

Its numbers. Roof 32 mm wide at 30 degrees, apex 10 mm up the slope from the pin, lips 5 mm, strips 3.5 mm, 9 mm high, 1.5 mm off the board on the pin's boss; it leans 12 degrees either way to its stop pegs. It weighs 2.5 g printed solid, 3.6 g with its steel ball.

The pin's friction. The rocker lies on the boss round its pin with its weight across the board, 3.4 g-force; PLA on PLA (0.3 to 0.4) at the boss's mean radius (about 2.9 mm) gives about 0.4 kg cm^2/s^2 (0.04 mN m), the pin itself a tenth of that. The simulation uses 0.4. At that, the rocker's own weight cannot turn it against the pin when it is near its stop (about 0.03 at 8 degrees), so it stays where the ball leaves it, 8 to 12 degrees over: that is still far enough for the next ball to choose its side.

The rocker alone (`tests/toggle.lua`, 16 balls each; the least lean any ball left it at):

	Passed	Least lean
as built	yes	9.1 degrees
pin friction 0, 1, 2, 4, 5	yes	11.6, 11.0, 10.3, 3.8, 0.5
pin friction 6	**no**: the rocker stops in the middle	1.3
friction 0.1 to 0.8	yes	7.8 to 9.8
slope 6 to 30 degrees	yes	9.6 to 11.6
run 5, 10, 15 cm longer (faster balls)	yes	7.6, 6.0, 2.8
run 20 cm longer	**no**: the ball comes too fast and jumps the cup	
let go up to 0.75 mm off centre (all the channel allows)	yes	8.7
tilted across 2 or 3 degrees one way, 2 the other	yes	7.1 to 10.0
tilted across 3 degrees the other way, or 4	**no**	
1200 and 4800 steps a second	yes	8.8, 9.9

Tilted across, the ball runs down its channel pressed against the downhill wall, and the board's sideways pull carries it past the apex onto the low side, which it then rolls down without turning the rocker. So the board must be level across: within 2 degrees.

The counter

The staircase. Each flip-flop 3.5 cm left of and 8 cm below the last, so a carry leaving the left cup falls straight into the next one's funnel; each done ball leaves the right cup into a lane straight down to the bins. The lanes' mouths were first 2.4 cm right of the pin: a ball came out of the cup and stuck, wedged between the rocker's lip and the lane's flared wall (found with `DC_CONTACTS`). At 2.8 cm they clear.

The feeder. A wheel of four notches turning under the channel of balls. Three things were found:

- A wall's box runs 1.5 mm past its ends (so walls meet solidly), and the ends of the channels at the wheel's housing reached into the wheel's path: the wheel rubbed them and stuck at friction 0.5. Those channels now start 2 mm further out.
- The hand must be able to push: seven waiting balls press on the wheel's rim. The simulation's hand turns the knob with up to 50 kg cm^2/s^2 (5 mN m, a light finger); it also passed with 20 and 1000.
- In the first wheel the teeth only touched the hub, which would have printed as five pieces: the hub is now 0.5 mm wider than the notches, so the wheel is one solid.

The tour, eight balls from 0, each reading and each ball's bin checked, the machine as built (pin friction 0.4), passed in every one of these:

- friction 0.1, 0.2, 0.3, 0.4, 0.5, 0.6
- pin friction 0, 1, 2, 3 (eight times the estimate)
- slope 8, 10, 15, 20, 25 degrees
- 1200, 2400, 4800 steps a second
- tilted across 1 and 2 degrees either way
- friction 0.6 with: tilted 2 degrees either way, slope 10 degrees, pin friction 2; friction 0.1 with: tilted 1 degree either way or 2 one way, slope 10 degrees, no pin friction
- the hand's torque 20 and 1000

and failed at pin friction 4 (a rocker stopped near the middle), tilted 3 degrees across either way, and at friction 0.1 tilted 2 degrees the other way. So: level across within 2 degrees, and within 1 to be sure.

What it costs

Little: eight balls, three rockers and the wheel, against some fifty fixed walls and pegs. The tour, about 28 simulated seconds, took 7 s of computing headless here; the exhibit rests (stops stepping) once nothing has moved for half a second.

What is idealised

- The knob is turned by the hand: a motor on the wheel's pin, exactly a quarter turn at a time. In the machine as built you turn it by its four marks (how far off a quarter it may be was not tried).
- The pins are hinges with a constant friction torque.
- Setting it back puts the balls and the rockers in place by hand.